Similarity & Proportional Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(5)$.

Final Answer: 38. Substitute $x = 5$: $f(5) = 6(5) + 8 = 38$.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A data set recorded for dilations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

4 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 45^\circ) = 84^\circ$.

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

On the enhanced ACT Math test (45 questions in 50 minutes), a student allocates 10 minutes to the first 15 questions and 16 minutes to questions 16 through 30. How many seconds per question remain for the final 15 questions, assuming 3 minutes are reserved at the end to check answers?

Time spent so far $= 10 + 16 = 26$ minutes. Leaving 3 minutes for review leaves $50 - 26 - 3 = 21$ minutes for the final 15 questions. Time per question $= \frac{21 \times 60 \text{ seconds}}{15 \text{ questions}} = \frac{1260}{15} = 84$ seconds per question.

6 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 1 · FoundationsThe Digital SAT Math section allocates 70 minutes across two modules of equal length. If each module contains 22 questions, what is the exact average time available per question in seconds?

Step 1: Calculate total seconds per module: $35 \text{ minutes} \times 60 \text{ s/min} = 2100 \text{ seconds}$. Step 2: Divide by the 22 questions in the module: $\frac{2100 \text{ seconds}}{22 \text{ questions}} = \frac{1050}{11} \approx 95.45$ seconds (about 1 min 35 s per question).

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple of 6 and 8.

Prime Factorization Method: $6 = 2^1 \times 3^1$ $8 = 2^3$ The Least Common Multiple (LCM) takes the highest power of every prime factor present: $\text{LCM} = 2^3 \times 3^1 = 8 \times 3 = \mathbf{24}$ ACT Tip: List multiples of the larger number until divisible by the smaller: $8, 16, 24$ (divisible by 6).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

Every Digital SAT question carries equal raw weight within a module. The student should immediately flag Questions 19–20, solve Questions 21–22 in under 90 seconds using Desmos, and spend the remaining 7.5 minutes on Questions 19–20 without time pressure.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

The Digital SAT SPR allows up to 5 characters including negative signs and decimal points:As a fraction: -7/4 (4 characters).As a terminating decimal: -1.75 (5 characters).Invalid: Mixed number -1 3/4 is read as -13/4 and will be graded incorrect.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Practice 1.2: If a and b are positive integers such that the greatest common factor of a and b is 6, and the least common multiple of a and b is 72, what is the product $a \times b$?

By the fundamental identity connecting greatest common factor and least common multiple: $\text{GCF}(a, b) \times \text{LCM}(a, b) = a \times b$. Substitute the given values: $a \times b = 6 \times 72 = 432$. The product is 432.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Practice 2.2: What is the sum of the solutions of the quadratic equation $3x^2 - 12x - 5 = 0$?

By Vieta's formulas, for any quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is $-\frac{b}{a}$. Here $a = 3$ and $b = -12$, so the sum of the solutions is: $-\frac{-12}{3} = \frac{12}{3} = 4$.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumA study of 100 students recorded test results based on exam prep: • Studied: 54 passed, 6 failed (60 total) • Did not study: 18 passed, 22 failed (40 total) What is the probability that a randomly chosen student passed the exam, given that the student studied?

The condition 'given that the student studied' restricts the denominator to the 60 students who studied: $P(\text{Passed} \mid \text{Studied}) = \frac{54}{60} = 0.90 = 90\%$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Why should you test Choice C (or H) first when back-solving numerical answer choices on the ACT?

ACT numerical choices are ordered from least to greatest (or greatest to least). Testing the middle choice tells you whether to move higher or lower, requiring at most 2 tests.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the greatest common factor (GCF) and least common multiple (LCM) of 24 and 36?

$24 = 2^3 \cdot 3$ and $36 = 2^2 \cdot 3^2$. Thus GCF $= 2^2 \cdot 3 =$ **12** and LCM $= 2^3 \cdot 3^2 =$ **72**.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

Practice 2.1: For what values of x is $|2x - 3| < 7$?

Rewrite as a compound inequality: $-7 < 2x - 3 < 7$. Add 3: $-4 < 2x < 10$. Divide by 2: $-2 < x < 5$. Solution set: $(-2, 5)$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Foundations & Reasoning

What is the sum and product of the roots of $3x^2 - 15x + 7 = 0$?

By Vieta's formulas, sum $= -b/a = 15/3 =$ **5** and product $= c/a =$ **7/3**.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Coordinate Geometry

For what values of x does $|2x - 5| > 9$ hold?

Split into two rays: $2x - 5 > 9 \Rightarrow x > 7$ or $2x - 57$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability

The equation of a circle in the xy-plane is $(x-5)^2+(y+2)^2=81$. What is the radius of the circle?

In standard circle form $(x-h)^2+(y-k)^2=r^2$, the constant on the right side represents r^2. Thus $r^2=81 \Rightarrow r=9$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Lines & Angles

A right triangle has legs 9 and 12. Find the hypotenuse.

Apply the Pythagorean Theorem $a^2+b^2=c^2$: $c=\sqrt{9^2+12^2}=\sqrt{81+144}=\sqrt{225}=\mathbf{15}$ ACT Scaling Shortcut: Notice that both legs are multiples of 3: $9=3\times3$ and $12=3\times4$. This is a $3-4-5$ Pythagorean triple scaled by factor 3: $3\times5=\mathbf{15}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Triangles & Congruence

A triangle has angles 50° and 60°. Find the third.

By the Triangle Angle Sum Theorem, the interior angles of any planar triangle sum to 180°: $\theta_1+\theta_2+\theta_3=180^\circ\ 50^\circ+60^\circ+\theta_3=180^\circ \Rightarrow 110^\circ+\theta_3=180^\circ\ \theta_3=180^\circ-110^\circ=\mathbf{70}^\circ$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x)=4x+6$ in translations, evaluate $f(2)$.

a. 14 ✓

b. 18

c. 8

d. 18

Substitute $x=2$: $f(2)=4(2)+6=14$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(2)$.

a. 17 ✓

b. 22

c. 10

d. 21

Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for translations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for rotations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in compositions of transformations, evaluate $f(3)$.

a. 16 ✓

b. 20

c. 12

d. 20

Substitute $x = 3$: $f(3) = 4(3) + 4 = 16$.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the figure, right triangle ABC has a right angle at B, and altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$. The altitude divides the hypotenuse into segments of lengths $AD = 9$ and $DC = 16$. Two similar right-triangular prisms, $\mathcal{P}_1$ and $\mathcal{P}_2$, have bases ΔABD and ΔBCD, respectively. If the volume of prism $\mathcal{P}_1$ is $135\ \text{cm}^3$, what is the volume of prism $\mathcal{P}_2$, in cm^3?

a. 320 ✓

b. 240

c. 625

d. 180

1. **Find the altitude and side lengths using the Geometric Mean Theorem:** In right triangle ABC with altitude $\overline{BD}$ to hypotenuse $\overline{AC}$: $BD = \sqrt{AD \cdot DC} = \sqrt{9 \cdot 16} = \sqrt{144} = 12$ Alternatively, we can find the hypotenuse $AC = AD + DC = 9 + 16 = 25$, and the legs of ΔABC using the Leg Geometric Mean Theorem: $AB = \sqrt{AD \cdot AC} = \sqrt{9 \cdot 25} = 15$ $BC = \sqrt{DC \cdot AC} = \sqrt{16 \cdot 25} = 20$ 2. **Determine the linear scale factor k between.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the diagram below, line segment DE is parallel to line segment BC. If $AD = 6$ cm, $DB = 4$ cm, and $AE = 9$ cm, what is the length of EC?

a. 6 cm

b. 15 cm

c. 13.5 cm ✓

d. 54 cm

Because line segment DE is parallel to line segment BC, by the Side-Splitter Theorem (also known as Thales's Theorem), triangle ADE is similar to triangle ABC. This similarity implies that the ratios of corresponding sides are equal. Specifically, we have the proportion: $\frac{AD}{DB} = \frac{AE}{EC}$ We are given $AD = 6$ cm, $DB = 4$ cm, and $AE = 9$ cm. Let $EC = x$. Substituting the given values into the proportion: $\frac{6}{4} = \frac{9}{x}$ To solve for x, we can cross-multiply: $6x = 4 \times 9$.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the figure shown, line segments AC and BD intersect at point E, and $\overline{AB} \parallel \overline{CD}$. If $AB = 9$, $CD = 15$, and $AE = 6$, what is the length of segment EC?

a. 3.6

b. 9

c. 10 ✓

d. 12

Because segment $\overline{AB}$ is parallel to segment $\overline{CD}$, alternate interior angles formed by transversals AC and BD are congruent: $\angle EAB \cong \angle ECD$ and $\angle EBA \cong \angle EDC$ Additionally, vertical angles $\angle AEB$ and $\angle CED$ are congruent. Therefore, $\Delta ABE \sim \Delta CDE$ by Angle-Angle (AA) similarity. Corresponding side lengths of similar triangles are proportional: $\frac{AB}{CD} = \frac{AE}{EC}$ Substitute the given values into the pro.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the converse of the statement: "If A triangle is equilateral, then it is equiangular."?

a. If A triangle is not equilateral, then it is not equiangular.

b. If It is not equiangular, then a triangle is not equilateral.

c. **If It is equiangular, then a triangle is equilateral.** ✓

d. If It is equiangular, then a triangle is not equilateral.

The converse swaps the hypothesis and conclusion of the original conditional.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the following conditional statement where the universe of discourse is the set of all parallelograms: x Which of the following correctly classifies the truth values of Statement P, its converse, its inverse, and its contrapositive, along with a mathematically valid counterexample for any false statement(s)?

a. Statement P is False (counterexample: a rhombus with unequal diagonal lengths); Converse is True; Inverse is True; Contrapositive is False (counterexample: a rhombus with unequal diagonal lengths). ✓

b. Statement P is False (counterexample: a rectangle with unequal adjacent sides); Converse is True; Inverse is True; Contrapositive is False (counterexample: a rectangle with unequal adjacent sides).

c. Statement P is False (counterexample: a rhombus with unequal diagonal lengths); Converse is False (counterexample: a non-square rhombus); Inverse is False (counterexample: a non-square rectangle); Contrapositive is False (counterexample: a rhombus with unequal diagonal lengths).

d. Statement P is True; Converse is False (counterexample: a rhombus with unequal diagonal lengths); Inverse is False (counterexample: a rhombus with unequal diagonal lengths); Contrapositive is True.

Let the universe of discourse be the set of all parallelograms. Define the component propositions: - p: "The diagonals of the parallelogram are perpendicular." - q: "The parallelogram is a square." 1. **Statement P (Conditional $p \to q$):** - A conditional statement $p \to q$ is false if and only if there exists a counterexample where the hypothesis p is true and the conclusion q is false. - A rhombus with unequal diagonal lengths (a non-square rhombus) is a parallelogram whose diagonals.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Point P lies on the directed line segment AB such that AP:PB = 2:3. Given the coordinates of A as $(-2, 5)$ and the coordinates of B as $(8, -5)$, find the coordinates of point P.

a. $(2, 2)$ ✓

b. $(4, -1)$

c. $(0, 0)$

d. $(6, -3)$

Let the coordinates of point A be $(x_1, y_1) = (-2, 5)$ and the coordinates of point B be $(x_2, y_2) = (8, -5)$. The point P partitions the directed line segment AB in the ratio m:n = 2:3. The formula for the coordinates of P (x, y) is given by: $x = \frac{nx_1 + mx_2}{m+n}$ $y = \frac{ny_1 + my_2}{m+n}$ Substitute the given values: $x = \frac{3(-2)+2(8)}{2+3} = \frac{-6+16}{5} = \frac{10}{5} = 2$ $y = \frac{3(5)+2(-5)}{2+3} = \frac{15-10}{5} = \frac{5}{5} = 1$ Therefore, the coordinates of point P are.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the statement: "If a quadrilateral has four right angles, then it is a square." ; ; Which of the following represents the inverse of this statement, and what is its truth value? ; ; Statement P: A quadrilateral has four right angles. ; Statement Q: It is a square.

a. If a quadrilateral is not a square, then it does not have four right angles. (True)

b. **If a quadrilateral does not have four right angles, then it is not a square. (True) ✓**

c. If a quadrilateral is not a square, then it does not have four right angles. (False)

d. If a quadrilateral does not have four right angles, then it is not a square. (False)

The original statement is of the form "If P, then Q." ; P: A quadrilateral has four right angles. ; Q: It is a square. ; ; The inverse of a conditional statement is formed by negating both the hypothesis and the conclusion. The inverse statement is: "If not P, then not Q." ; Not P: A quadrilateral does not have four right angles. ; Not Q: It is not a square. ; ; Therefore, the inverse statement is: "If a quadrilateral does not have four right angles, then it is not a square." ; ; To determine the truth value of the.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Point M is the midpoint of segment $\overline{AB}$. If $AM = 3x + 1$ and $MB = 5x - 9$, what is the length of segment $\overline{AB}$?

a. 5

b. 16

c. 26

d. **32 ✓**

1. **Apply the definition of a midpoint:** Since M is the midpoint of segment $\overline{AB}$, the two sub-segments have equal lengths: $AM = MB$ Substitute the given algebraic expressions: $3x + 1 = 5x - 9$ 2. **Solve for x:** Subtract $3x$ from both sides: $1 = 2x - 9$ Add 9 to both sides: $10 = 2x \Rightarrow x = 5$ 3. **Calculate the lengths of the sub-segments:** $AM = 3(5) + 1 = 16$ $MB = 5(5) - 9 = 16$ 4. **Apply the Segment Addition Postulate:** The total length of $\overline{AB}$ is the sum of the pa.

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Two vertical poles of heights 10 m and 15 m stand on level ground. Guy wires connect the top of each pole to the base of the other pole. At what height above the ground do the two crossing wires intersect?

a. 7.5 m

b. 5 m

c. **6 m ✓**

d. 6.25 m

Let the poles be $a = 10$ and $b = 15$. By setting up proportions in the two pairs of similar triangles formed by the crossing wires and the ground, the height h of the intersection satisfies the crossed-ladders formula: $\frac{1}{h} = \frac{1}{a} + \frac{1}{b}$. Substituting: $\frac{1}{h} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6} \Rightarrow h = 6$ m. Remarkable fact: This height is completely independent of the distance between the poles!

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

To determine the height of a sea cliff, a surveyor places a small mirror flat on horizontal ground 18 meters from the base of the cliff. The surveyor steps back 2 meters from the mirror until the top of the cliff is aligned with the center of the mirror. If the surveyor's eye level is 1.6 meters above the ground, what is the height of the cliff?

a. 16.2 m

b. 22.5 m

c. **14.4 m ✓**

d. 18.0 m

By the law of optical reflection, the angle of incidence equals the angle of reflection. Both the surveyor and the cliff stand perpendicular to the ground (90°). By AA similarity:
$\frac{\text{Cliff Height}}{\text{Distance to Mirror}} = \frac{\text{Eye Height}}{\text{Surveyor Distance to Mirror}} \Rightarrow \frac{H}{18} = \frac{1.6}{2} = 0.8$. Multiplying gives $H = 18 \times 0.8 = 14.4$ m.

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A pinhole camera has its pinhole at point E. An object AB of height 15 cm is placed 45 cm in front of E. The inverted image CD forms on a screen 30 cm behind E. Assuming rays travel in straight lines intersecting at E parallel to the camera plates, what is the image height CD?

a. **10 cm ✓**

b. 12 cm

c. 8 cm

d. 7.5 cm

Light rays passing through the pinhole form similar triangles on either side of E: $\Delta ABE \sim \Delta CDE$. The ratio of object height to image height equals the ratio of distances from the pinhole:
$\frac{CD}{AB} = \frac{d_2}{d_1} \Rightarrow \frac{CD}{15} = \frac{30}{45} = \frac{2}{3} \Rightarrow CD = 15 \times \frac{2}{3} = 10$ cm.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In ΔABC, segments $DE \parallel FG \parallel BC$ with D, F on AB and E, G on AC. If $AD = 2$, $DF = 3$, $FB = 5$, and base $BC = 30$, what is the difference between the lengths of segments FG and DE?

a. 6

b. **9** ✓

c. 12

d. 15

Total length $AB = 2 + 3 + 5 = 10$. For $DE \parallel BC$, the scale factor from A is $\frac{AD}{AB} = \frac{2}{10} = 0.2 \Rightarrow DE = 30 \times 0.2 = 6$. For $FG \parallel BC$, length $AF = AD + DF = 2 + 3 = 5$. The scale factor is $\frac{AF}{AB} = \frac{5}{10} = 0.5 \Rightarrow FG = 30 \times 0.5 = 15$. The difference is $FG - DE = 15 - 6 = 9$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability

A 6-foot-tall person stands 10 feet away from the base of a 16-foot streetlight. How long is the person's shadow cast along the ground?

a. 4 ft

b. 8 ft

c. 5 ft

d. **6 ft** ✓

Let s be the length of the person's shadow. The light source creates nested similar right triangles: the small triangle (person and shadow) and the large triangle (streetlight and total distance): $\frac{16}{10+s} = \frac{6}{s}$. Cross-multiplying: $16s = 6(10+s) \Rightarrow 16s = 60 + 6s \Rightarrow 10s = 60 \Rightarrow s = 6$ ft.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Lines & Angles

From an exterior point P, two secant lines are drawn to a circle. The first secant intersects the circle at A and B with $PA = 4$ and $AB = 8$. The second secant intersects the circle at C and D with $PC = 3$. What is the length of chord CD?

a. 16

b. 12

c. **13** ✓

d. 15

For the first secant, the entire secant length is $PB = PA + AB = 4 + 8 = 12$. By the Secant-Secant Power Theorem (derived from similar triangles $\Delta PAD \sim \Delta PCB$): $PA \cdot PB = PC \cdot PD$. Substituting: $4 \times 12 = 3 \cdot PD \Rightarrow 48 = 3 \cdot PD \Rightarrow PD = 16$. The chord length is $CD = PD - PC = 16 - 3 = 13$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Triangles & Congruence

Two isosceles triangles each have a vertex angle measuring 40°. The leg of the smaller triangle has length 15, and the corresponding leg of the larger triangle has length 25. If the base of the smaller triangle is 12, what is the base of the larger triangle?

a. 18

b. 24

c. **20** ✓

d. 22

In both isosceles triangles, the base angles are equal: $\frac{180^\circ - 40^\circ}{2} = 70^\circ$. Because all three angles match ($40^\circ, 70^\circ, 70^\circ$), the triangles are similar by AA. The scale factor is $k = \frac{25}{15} = \frac{5}{3}$. The base of the larger triangle is $12 \times \frac{5}{3} = 20$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

The sides of ΔABC measure 7, 10, and 12. A similar triangle ΔDEF has a shortest side of length 21. What is the perimeter of ΔDEF?

a. 58

b. **87** ✓

c. 72

d. 96

The shortest side of ΔABC is 7. The corresponding shortest side of ΔDEF is 21. The linear scale factor is $k = \frac{21}{7} = 3$. The perimeter of ΔABC is $P_1 = 7 + 10 + 12 = 29$. Since perimeters scale linearly by k, the perimeter of ΔDEF is $P_2 = 3 \times 29 = 87$.

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42 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 6.01 Polygons & Quadrilaterals**

In ΔABC, $\angle A = 42^\circ$ and $\angle B = 68^\circ$. In ΔDEF, $\angle D = 70^\circ$ and $\angle E = 68^\circ$. Which statement correctly determines whether the triangles are similar?

a. **The triangles are similar by AA because $\angle C = 70^\circ$, matching $\angle D$** ✓

b. The triangles are not similar because $\angle A \not\cong \angle D$

c. The triangles are similar by SAS using sides adjacent to 68°

d. Similarity cannot be determined without side lengths

In ΔABC, the interior angles sum to 180°: $\angle C = 180^\circ - 42^\circ - 68^\circ = 70^\circ$. In ΔDEF, two angles are 68° and 70°. Both triangles share angles of 68° and 70°. By the Angle-Angle (AA) Similarity Postulate, $\Delta ABC \sim \Delta FED$.

43 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 7.01 Circles**

Chords AB and CD intersect inside a circle at point P. If $PA = 4$, $PB = 9$, and $PC = 6$, what is the length of segment PD?

a. **6** ✓

b. 8

c. 5.4

d. 7

Triangles ΔPAC and ΔPDB are similar by AA (vertical angles at P and inscribed angles intercepting arc BC). Corresponding sides yield the Intersecting Chords Theorem: $PA \cdot PB = PC \cdot PD$. Substituting: $4 \times 9 = 6 \cdot PD \Rightarrow 36 = 6 \cdot PD \Rightarrow PD = 6$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Which of the following conditions is NOT sufficient to prove that two triangles are similar?

a. Two pairs of corresponding angles are congruent (AA)

b. Three pairs of corresponding sides are proportional (SSS)

c. **Two pairs of sides are proportional and a non-included angle is congruent (SSA) ✓**

d. Two pairs of sides are proportional and their included angle is congruent (SAS)

SSA (Side-Side-Angle) is NOT a valid similarity criterion. When the congruent angle is not included between the proportional sides, the third side can swing to form two non-congruent, non-similar triangles (the ambiguous case). In contrast, AA, SAS, and SSS all rigorously guarantee triangle similarity.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Transformations & Symmetry

A cone with height 30 cm is sliced by a plane parallel to its base at a distance of 15 cm from the apex, creating a smaller top cone and a bottom frustum. What fraction of the original cone's volume is contained in the frustum?

a. 1/2

b. 3/4

c. 1/8

d. **7/8 ✓**

The smaller top cone is similar to the original cone with linear scale factor $k = \frac{15}{30} = \frac{1}{2}$. The volume of the smaller cone is $k^3 = (1/2)^3 = \frac{1}{8}$ of the original cone's volume. The remaining frustum contains $1 - \frac{1}{8} = \frac{7}{8}$ of the original volume.

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